

## 10 > Secure segmentation, guest access & remote work

```
edit "VOIP-HIGH"
  set guaranteed-bandwidth 15000
  set priority high
  set dscp-match 46
next
edit "PAYROLL-MID"
  set guaranteed-bandwidth 8000
  set priority medium
  set protocol 6
  set dst-port 443
next
end
```

### B. Build a shaping policy

```
config firewall shaping-policy
  edit 1
    set srcintf "lan"
    set dstintf "wan1"
    set traffic-shaper "VOIP-HIGH"
    set dstaddr "all"
    set service "ALL"
    set schedule "always"
  next
end
```

### C. Verify

CLI: `diagnose sys qos stat` – VOIP should show zero drops during calls.

GUI: Policy & Objects > Traffic Shapers graphs live utilisation.

### Payroll protection drill

1. Start a Teams video call; run `iperf3 -u -b 150M` from another PC to flood link.
2. Without QoS, call latency spikes >400 ms. With the above policy, FortiGate guarantees 15 Mbps; latency stays <80 ms – proven in the community test bench.

**Tip:** On Cisco IOS, enable auto qos voip trust on access ports for IP phones; it builds DSCP maps in one command.

## 3) Setting up a secure guest Wi-Fi

### Why offer guest Wi-Fi at all?

Whether you're a two-person design studio in Pune or a 40-seat clinic in Coimbatore, visitors now expect connectivity. A dedicated guest SSID keeps strangers off your corporate VLAN while still giving them pipes to the public Internet. In fact, Cisco research shows that cafés and retail outlets see a 35% increase in dwell time once reliable guest Wi-Fi is advertised.

### Core design goals

- **Isolation:** Guests must never traverse internal subnets.

- **Accountability:** A log tying devices to phone numbers or OTPs satisfies India's DoT public-Wi-Fi rules.
- **Throttle & filter:** Cap bandwidth to stop Netflix hogging and block questionable domains.



### Ten-Minute lab – Wizard-driven captive portal (Cisco WAP-150)

**Hardware:** Any Cisco Small Business AP with firmware ≥ 1.0.2.8.

**Outcome:** A WPA2-PSK guest SSID with web-auth splash page and 2 Mb/s cap.

1. **Run the Setup Wizard:** Dashboard ► Run Setup Wizard.
2. **Enable 'Guest Access (Captive Portal)'** when prompted, name the SSID "YourBiz-Guest" and tick Hide private SSID.
3. **Security Type:** Choose WPA2-Personal; enter an easy-to-dictate pass-phrase, e.g. 22@YourBiz2025.
4. **Custom Splash Page:** Paste your logo and add Terms of Use – include a no-torrent clause to stay inside India's IT Act safe-harbour.
5. **Bandwidth Limiter:** Under Advanced ► Traffic Shaping set Ingress/Egress = 2 Mbps.
6. **DHCP Scope:** In LAN ► VLAN create VLAN 40, subnet 192.168.40.0/24; mark the uplink as trunk, tagged 40.
7. **Firewall ACL:** On the edge router, deny VLAN 40 → Corp-LAN (/10, /20); permit VLAN 40 → Internet only.
8. **Test:** Connect a phone, load `192.168.40.1`, accept splash, browse; then `ping 192.168.10.10` (finance server) should fail.

### Hardening Checklist

| Risk                            | Mitigation                                                  |
|---------------------------------|-------------------------------------------------------------|
| Rogue AP broadcasting same SSID | Enable 802.11w Management Frame Protection                  |
| Splash-bypass via static IP     | ACL blocks RFC 1918 to Corp subnets                         |
| Legal traceability              | SMS OTP or phone-based voucher complies with DoT Café rules |